

Plan 2 / Route δ : Concrete Proof Program for Sharp Quantitative Faber–Krahn Stability

Abstract

This note records the current concrete route from the Plan 2 level-set machinery to the sharp quantitative Faber–Krahn inequality. It is meant to be a compact map of the proof, not a replacement for the individual building blocks. The load-bearing chain is now:

LEVELSETIDENTITY $\rightarrow$ LIMSUPDEFORMATION $\rightarrow$ SAMECENTREFJ $\rightarrow$ TRIMMEDVELOCITYREPAIR
 $\rightarrow$ WEIGHTEDMETRICTRACE $\rightarrow$ GOODENDPOINTTRACE $\rightarrow$ BOUNDARYLAYERTRANSFER $\rightarrow$ GLOBALASSEMBLY.

Three previously conditional links are now closed. The conditional METRICFRAMEWORK hypothesis (upper-gradient recovery) is discharged by an unconditional limsup deformation estimate (LIMSUPDEFORMATION); the conditional same-centre Fusco–Julin hypothesis is discharged by a joint-functional theorem (SAMECENTREFJ); and the exact endpoint ρ_δ at which the metric trace is anchored can be replaced by any Borel good level within $O(\delta_T)$ of ρ_δ (GOODENDPOINTTRACE), which preserves the sharp exponent and decouples the trace from the exact endpoint. The centroid and space-time notes remain useful diagnostics, but they are not part of the shortest current proof. The former pointwise lower gradient hypothesis Hyp-G is also no longer a standing assumption: on Fusco–Julin good levels, the low-gradient part of the level surface is self-truncated and paid for by the Cauchy–Schwarz defect D_H . The remaining *honest* dependencies are the two bad-level kinetic action bounds in [7, Hyp. 3.6]; these were stated in [7] as conditional inputs and remain so.

1 Target and normalisation

The final theorem is the sharp quantitative Faber–Krahn inequality: there is $c_{\text{FK}}(n) > 0$ such that every open $\Omega \subset \mathbb{R}^n$, $0 < |\Omega| < \infty$, satisfies

$$\left(\frac{|\Omega|}{|B_1|} \right)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(n) \mathcal{A}(\Omega)^2, \quad (1)$$

where $|B^*| = |\Omega|$ and

$$\mathcal{A}(\Omega) := \inf_{x_0 \in \mathbb{R}^n} \frac{|\Omega \Delta (x_0 + B^*)|}{|B^*|}.$$

The exponent 2 is sharp, as shown by smooth volume-preserving ellipsoidal perturbations of the ball.

The Plan 2 core proves first the bounded Saint–Venant stability

$$\mathcal{A}(\Omega)^2 \leq C(n, R) \delta_T(\Omega), \quad \delta_T(\Omega) := E(\Omega) - E(B_1), \quad (2)$$

for $|\Omega| = \omega_n$ and $\Omega \subset B_R$, where $E(\Omega) = -T(\Omega)/2$ and $T(\Omega)$ is torsional rigidity. The global Faber–Krahn statement then follows from the BDV bounded reduction and the Kohler–Jobin transfer, exactly as in the Plan 1 final library.

Throughout the bounded part we fix $|\Omega| = \omega_n$ and write $u = u_\Omega$ for the torsion function, $-\Delta u = 1$ in Ω , $u \in H_0^1(\Omega)$. For almost every level,

$$E_\rho := \{u > t(\rho)\}, \quad |E_\rho| = \omega_n \rho^n, \quad d\mu(\rho) := (-t'(\rho)) d\rho.$$

The endpoint used in the final transfer is

$$\rho_\delta := 1 - \kappa \sqrt{\delta_T(\Omega)}. \quad (3)$$

For small deficit, ρ_δ lies in a fixed outer collar $[\rho_c, 1]$, for instance with $\rho_c = 3/4$. This collar localisation is enough for the argument.

2 Step 1: exact level-set deficit identity

The first input is the finite-perimeter identity in `LEVELSETIDENTITY`. For $E_t = \{u > t\}$, $m(t) = |E_t|$, define

$$\begin{aligned} \alpha(t) &:= \int_{\partial^* E_t} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1}, & \beta(t) &:= \int_{\partial^* E_t} |\nabla u| d\mathcal{H}^{n-1}, \\ D_H(t) &:= \alpha(t)\beta(t) - P(E_t)^2, & D_I(t) &:= P(E_t)^2 - n^2 \omega_n^{2/n} m(t)^{2-2/n}. \end{aligned}$$

Then for every open set of finite measure,

$$\frac{2}{|\Omega|} (E(\Omega) - E(B)) = \frac{1}{|\Omega|} \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2 \omega_n^{2/n} m(t)^{1-2/n}} dt. \quad (4)$$

Both defects are nonnegative. The proof uses only BV coarea on reduced boundaries, Lipschitz truncation for the flux identity $\int_{\partial^* E_t} |\nabla u| = |E_t|$, and the corrected rearranged second derivative $I'' = -1/\alpha$.

On every fixed collar $\rho \in [\rho_c, \rho_\delta]$, (4) gives

$$\int_{\rho_c}^{\rho_\delta} (D_H(t(\rho)) + D_I(t(\rho))) d\mu(\rho) \leq C(n, \rho_c) \delta_T(\Omega). \quad (5)$$

We also use the Talenti comparison in two elementary forms: $-t'(\rho) \leq 1/n$, and a bad-set estimate

$$|\{\rho \in [\rho_c, \rho_\delta] : -t'(\rho) < \rho_c/(2n)\}| \leq C(n, \rho_c) \delta_T(\Omega).$$

The second estimate is the profile-gap input needed to pass from $d\mu$ -weighted kinetic energy to Lebesgue kinetic energy.

3 Step 2: metric first variation

Let $\mathcal{X}$ be the space of finite-measure L^1 classes modulo translations:

$$d_{\mathcal{X}}([A], [C]) = \inf_{a \in \mathbb{R}^n} |A \Delta (C + a)|.$$

Set $F_\rho := \rho^{-1} E_\rho$ so $|F_\rho| = \omega_n$. For a Borel centre field z_ρ , define

$$V_\rho(x) := \frac{-t'(\rho)}{|\nabla u(x)|}, \quad H_{z_\rho, \rho}(x) := \frac{(x - z_\rho) \cdot \nu_\rho(x)}{\rho} \quad (x \in \partial^* E_\rho).$$

The new building block `LIMSUPDEFORMATION` (Theorem 1.1, Plan 2 Deliverable A) proves the finite-perimeter metric first-variation estimate unconditionally:

$$|\dot{F}_\rho|_{\mathcal{X}} \leq \rho^{-n} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1} \quad (6)$$

for almost every $\rho \in [\rho_c, 1]$. The proof is a direct coarea computation on the symmetric difference $E_{\rho+h} \Delta T_h(E_\rho)$ with T_h the affine pull-back combining homothety and a chosen translation; no Reshetnyak or *BV* lower semicontinuity is used. The previous `METRICFRAMEWORK` formulation, which left (6) conditional on an upper-gradient recovery hypothesis, is therefore discharged.

4 Step 3: isoperimetric good levels

Fix a small dimensional threshold $\eta_I > 0$ and set

$$G_I := \{\rho \in [\rho_c, \rho_\delta] : D_I(t(\rho)) \leq \eta_I\}, \quad B_I := [\rho_c, \rho_\delta] \setminus G_I.$$

By (5),

$$\mu(B_I) \leq C \delta_T(\Omega). \quad (7)$$

On G_I , the building block SAMECENTREFJ (Plan 2 Deliverables B and C) proves the joint same-centre Fusco–Julin package unconditionally: there is one Borel centre field $\rho \mapsto z_\rho \in \overline{B_{R+1}}$ such that

$$d\mathcal{X}(F_\rho, B_1)^2 \leq C(n, R, \rho_c) D_I(t(\rho)). \quad (8)$$

The same-centre package, together with the finite-perimeter oriented radial divergence estimate proved in SAMECENTREFJ/FJCENTERBRIDGE, gives the homothetic trace bound

$$\left(\int_{\partial^* E_\rho} |H_{z_\rho, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C(n, R, \rho_c) D_I(t(\rho)), \quad (9)$$

where

$$\bar{V}_\rho := \frac{1}{P(E_\rho)} \int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1} = \frac{P(B_\rho)}{P(E_\rho)}.$$

At this stage the same-centre theorem and its measurable-selection input are unconditional: [5, Thm. 3.6 + Prop. 4.2]. They are no longer hidden inside WEIGHTEDMETRICTRACE and no longer conditional on an external citation.

5 Step 4: self-truncated velocity repair

This is the repair replacing the former lower-gradient hypothesis. On a good level write $S_\rho := \partial^* E_\rho$, $f := |\nabla u|$, $P := P(E_\rho)$, and

$$\beta := \int_{S_\rho} f d\mathcal{H}^{n-1} = |E_\rho|, \quad \bar{f} := \beta/P.$$

Split

$$S_\rho = G_\rho \cup L_\rho, \quad G_\rho := \{f \geq \bar{f}/2\}, \quad L_\rho := \{f < \bar{f}/2\}.$$

The weighted variance identity for D_H gives

$$\int_{L_\rho} \frac{1}{\bar{f}} d\mathcal{H}^{n-1} \leq \frac{4}{\beta} D_H(t(\rho)). \quad (10)$$

Thus the low-gradient part is small in the only measure relevant to normal velocity. On G_ρ , $f \geq \bar{f}/2$ gives the reciprocal variance comparison formerly supplied by a pointwise lower bound. The resulting estimate is

$$\left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C(n, \rho_c) D_H(t(\rho)), \quad \rho \in G_I. \quad (11)$$

This is the key point: tiny components, cusps, or other low-gradient pieces do not need uniform non-degeneracy. They are discarded at the level surface and charged to D_H itself. The discarded swept volume also satisfies

$$\int_{G_I} \int_{L_\rho} V_\rho d\mathcal{H}^{n-1} d\rho \leq C(n, \rho_c) \delta_T(\Omega).$$

6 Step 5: integrated action and kinetic energy

On G_I , choose the same-centre field z_ρ in (6), set $a = 0$, and use the triangle inequality:

$$|V_\rho - H_{z_\rho, \rho}| \leq |V_\rho - \bar{V}_\rho| + |H_{z_\rho, \rho} - \bar{V}_\rho|.$$

Together with (9) and (11), this gives

$$|\dot{F}_\rho|_{\mathcal{X}}^2 \leq C(n, R, \rho_c)(D_H(t(\rho)) + D_I(t(\rho))) \quad (\rho \in G_I). \quad (12)$$

On B_I , the repaired `WEIGHTEDMETRICTRACE` no longer invokes the unsupported perimeter-free Lipschitz bound (A0); instead it makes its bad-level kinetic requirements explicit as conditional hypotheses. Combining those hypotheses with (5) and (8) yields the integrated action estimate

$$\int_{\rho_c}^{\rho_\delta} \left(d_{\mathcal{X}}(F_\rho, B_1)^2 + |\dot{F}_\rho|_{\mathcal{X}}^2 \right) d\mu(\rho) \leq C(n, R, \rho_c) \delta_T(\Omega). \quad (13)$$

Using the Talenti bad-set estimate for $\{-t'(\rho) < \rho_c/(2n)\}$, the same bound implies the Lebesgue kinetic estimate

$$\int_{\rho_c}^{\rho_\delta} |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq C(n, R, \rho_c) \delta_T(\Omega). \quad (14)$$

7 Step 6: weighted metric trace at a good endpoint

Let $L^* := (\rho_\delta - \rho_c)/4$ and $J^* := [\rho_\delta - L^*, \rho_\delta]$. From (13) and the measure lower bound $\mu(J^*) \geq c(n, \rho_c)L^* - C\delta_T$, Markov gives a point $\rho_0 \in J^*$ with

$$d_{\mathcal{X}}(F_{\rho_0}, B_1)^2 \leq C\delta_T(\Omega).$$

For any $\rho \in J^*$, the metric absolute-continuity inequality and (14) give

$$d_{\mathcal{X}}(F_\rho, F_{\rho_0})^2 \leq |\rho - \rho_0| \int_{J^*} |\dot{F}_s|_{\mathcal{X}}^2 ds \leq C\delta_T(\Omega).$$

The triangle inequality yields the uniform window trace

$$\sup_{\rho \in J^*} d_{\mathcal{X}}(F_\rho, B_1)^2 \leq C(n, R, \rho_c) \delta_T(\Omega).$$

The new building block `GOODENDPOINTTRACE` (Plan 2 Deliverable D) then promotes this to a Borel endpoint

$$\hat{\rho} \in [\rho_\delta - C_1\delta_T, \rho_\delta] \cap G_I \cap G_\tau, \quad d_{\mathcal{X}}(F_{\hat{\rho}}, B_1)^2 \leq C(n, R, \rho_c) \delta_T(\Omega).$$

The existence of $\hat{\rho}$ uses the Lebesgue bad-set measure bounds $\mathcal{L}^1(B_\tau) \leq C\delta_T$ (profile-gap) and $\mathcal{L}^1(B_I \cap G_\tau) \leq C\delta_T$ (deficit identity over G_τ). This is exactly where the two defects interact: D_I supplies many levels close to balls, while D_H supplies the kinetic control needed to transport that closeness uniformly through the window. The substitution $\rho_\delta \rightsquigarrow \hat{\rho}$ feeds the downstream boundary-layer transfer without changing the sharp exponent (see Step 7).

8 Step 7: boundary-layer transfer from $\hat{\rho}$

By definition of $d_{\mathcal{X}}$ and the scaling $F_\rho = \rho^{-1}E_\rho$, the good-endpoint trace implies

$$\mathcal{A}(E_{\hat{\rho}})^2 \leq C(n, R, \rho_c) \delta_T(\Omega), \quad (15)$$

where the asymmetry of $E_{\hat{\rho}}$ is measured at its own volume $\omega_n \hat{\rho}^n$. Since $E_{\hat{\rho}} \subset \Omega$ and $\hat{\rho} \in [\rho_\delta - C_1 \delta_T, \rho_\delta]$,

$$|\Omega \setminus E_{\hat{\rho}}| = \omega_n(1 - \hat{\rho}^n) \leq \omega_n n(1 - \hat{\rho}) \leq \omega_n n(\kappa \sqrt{\delta_T} + C_1 \delta_T) \leq C(n) \sqrt{\delta_T(\Omega)}.$$

The elementary superlevel transfer lemma gives

$$\mathcal{A}(\Omega) \leq \mathcal{A}(E_{\hat{\rho}}) + \frac{2}{\omega_n} |\Omega \setminus E_{\hat{\rho}}|.$$

Squaring and using (15) gives the bounded sharp Saint–Venant inequality

$$\mathcal{A}(\Omega)^2 \leq C(n, R, \rho_c) \delta_T(\Omega) \quad (16)$$

in the small-deficit regime. The cross term $O(\delta_T^{3/2})$ from $\rho_\delta - \hat{\rho}$ is dominated by the leading $O(\sqrt{\delta_T})$ collar, so the sharp exponent is preserved. For large deficit the trivial bound $\mathcal{A}(\Omega) < 2$ is absorbed by increasing C .

9 Step 8: bounded reduction and Kohler–Jobin

Fix $\rho_c = 3/4$ for the Plan 2 core and then apply the bounded result at the BDV dimensional radius $R_*(n)$. The bounded-to-global reduction imported from Plan 1 gives

$$E(\Omega) - E(B_1) \geq c_{\text{SV}}^{\text{glob}}(n) \mathcal{A}(\Omega)^2 \quad (|\Omega| = \omega_n).$$

Scaling gives the corresponding global Saint–Venant estimate at arbitrary volume. The Kohler–Jobin transfer then yields (1) with

$$c_{\text{FK}}(n) = \min \left\{ \frac{4L_n c_{\text{SV}}^{\text{glob}}(n)}{(n+2)T_n}, \frac{L_n(2^{2/(n+2)} - 1)}{4} \right\},$$

where $L_n = \lambda_1(B_1)$ and $T_n = T(B_1)$.

10 What is load-bearing, and what is not

The load-bearing proof is the eight-step chain above. The current role of the surrounding WIP files is as follows.

File	Role	Status in the proof
<code>WIP_LevelSetIdentity</code>	exact $D_H + D_I$ identity	load-bearing
<code>WIP_LimsupDeformation</code>	metric first variation, unconditional	load-bearing
<code>WIP_SameCentreFJ</code>	joint same-centre FJ + Borel selection	load-bearing
<code>WIP_MetricFramework</code>	smooth identity + $\mathcal{X}$ setup	superseded by <code>LIMSUPDEFORMATION</code>
<code>WIP_FJCenterBridge</code>	oriented radial divergence trace	bridge, now unconditional
<code>WIP_TrimmedVelocityRepair</code>	removes Hyp-G(lower)	load-bearing
<code>WIP_WeightedMetricTrace</code>	action/kinetic/uniform trace	load-bearing modulo bad-kinetic
<code>WIP_GoodEndpointTrace</code>	Borel good endpoint $\hat{\rho}$	load-bearing
<code>WIP_BoundaryLayerTransfer</code>	endpoint back to Ω	load-bearing
<code>WIP_GlobalAssembly</code>	BDV reduction + KJ transfer	load-bearing
<code>WIP_CentroidBound</code>	centroid diagnostics	auxiliary
<code>WIP_SpaceTimeIdentity</code>	space-time diagnostics	auxiliary

In particular, the following are not used in the current shortest route: BDV/Cicalese–Leonardi selection, graph entry, Schauder interpolation, nearly-spherical expansions, pointwise radial trace (R), pointwise (B^2), and the pointwise lower-gradient Hyp-G.

11 Audit checklist

The remaining useful audit tasks are now sharply focused:

1. (*Discharged.*) The upper-gradient recovery theorem needed for (6) is replaced by the direct finite-perimeter limsup deformation estimate of [3, Thm. 1.1]. Audit: verify the Lebesgue-point input (genericity (iii) there) and the compact-strip device.
2. (*Discharged.*) The exact same-centre Fusco–Julin theorem and its measurable-selection input are now proved in [5, Thms. 3.6, 4.2]. Audit: verify the “centres-close” Theorem 3.4 against the Fuglede / Cicalese–Leonardi nearly-spherical inputs.
3. Check the self-truncated algebra (10)–(11); it is the active replacement for Hyp-G.
4. Verify the repaired Talenti profile-gap bad-set estimate used to pass from (13) to (14).
5. Verify the remaining bad-level kinetic hypotheses now stated in [7, Hyp. 3.6]. These are *not* discharged by the present round of Deliverables A–D; the GOODENDPOINTTRACE variant gives a cleaner downstream interface, but the underlying trace-uniform theorem still uses both bad-level action bounds.
6. Check the superlevel transfer from $E_{\hat{\rho}}$ to Ω and the constant bookkeeping in the BDV/Kohler–Jobin imports.

References

- [1] Plan 2 building block LEVELSETIDENTITY, `WIP_LevelSetIdentity.tex`.
- [2] Plan 2 building block METRICFRAMEWORK, `WIP_MetricFramework.tex`.
- [3] Plan 2 Deliverable A, LIMSUPDEFORMATION, `WIP_LimsupDeformation.tex`.
- [4] Plan 2 bridge block FJCENTERBRIDGE, `WIP_FJCenterBridge.tex`.
- [5] Plan 2 Deliverables B and C, SAMECENTREFJ, `WIP_SameCentreFJ.tex`.
- [6] Plan 2 building block TRIMMEDVELOCITYREPAIR, `WIP_TrimmedVelocityRepair.tex`.
- [7] Plan 2 building block WEIGHTEDMETRICTRACE, `WIP_WeightedMetricTrace.tex`.
- [8] Plan 2 Deliverable D, GOODENDPOINTTRACE, `WIP_GoodEndpointTrace.tex`.
- [9] Plan 2 building block BOUNDARYLAYERTRANSFER, `WIP_BoundaryLayerTransfer.tex`.
- [10] Plan 2 building block GLOBALASSEMBLY, `WIP_GlobalAssembly.tex`.
- [11] Plan 1 building block BOUNDEDREDUCTION, `Final/BoundedReduction.tex`.
- [12] Plan 1 building block KOHLERJOBINTRANSFER, `Final/KohlerJobinTransfer.tex`.